



Quantifying Socio-Linguistic Harm in Low-Resource Languages: A Conceptual Framework for Representational and Expressive Dimensions

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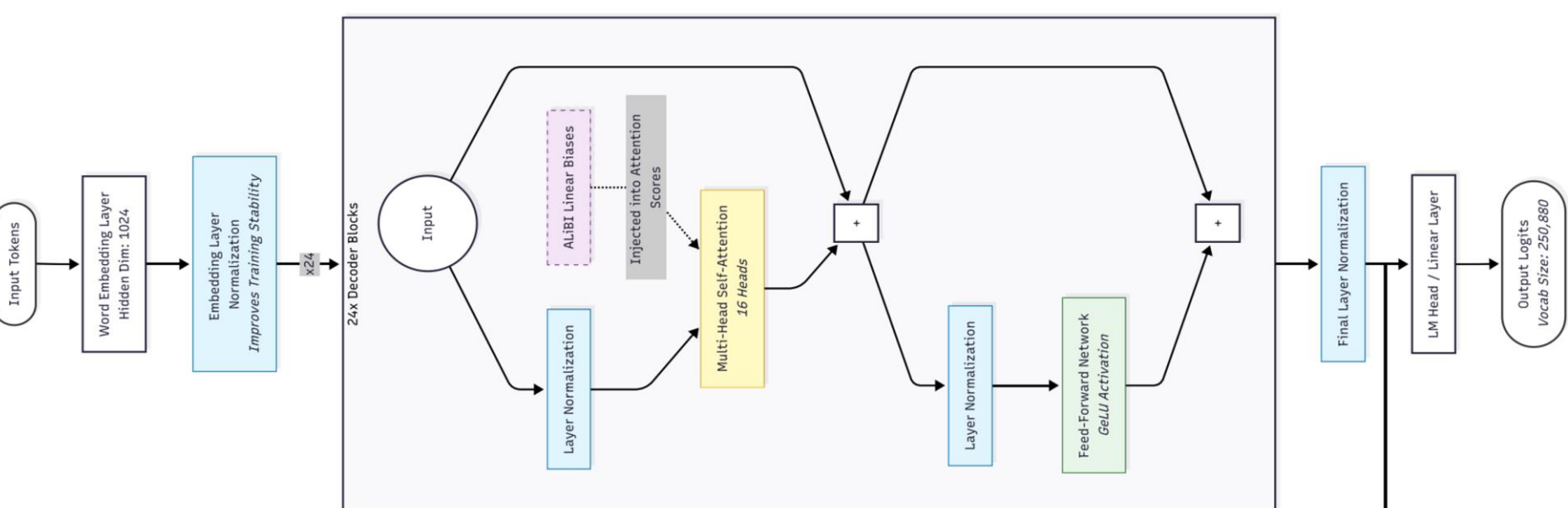
Introduction

Addressing harmful speech is vital because socio-linguistic discrimination inflicts severe psychological damage, reinforces systemic power imbalances, and silences marginalized groups.

The problems with current NLP systems are,

- **Reliance on Binaries:** Harmful speech detection is largely limited to binary labels (e.g., Hate vs. Non-hate) [1].
- **Missing the Nuance:** This fails to capture the interplay between what is said (Representational stance) and how it is conveyed (Expressive stance) [2].
- **Low-Resource Language Gap:** In morphologically rich languages like Tamil and Sinhala, this limitation is severe [3].

Motivation

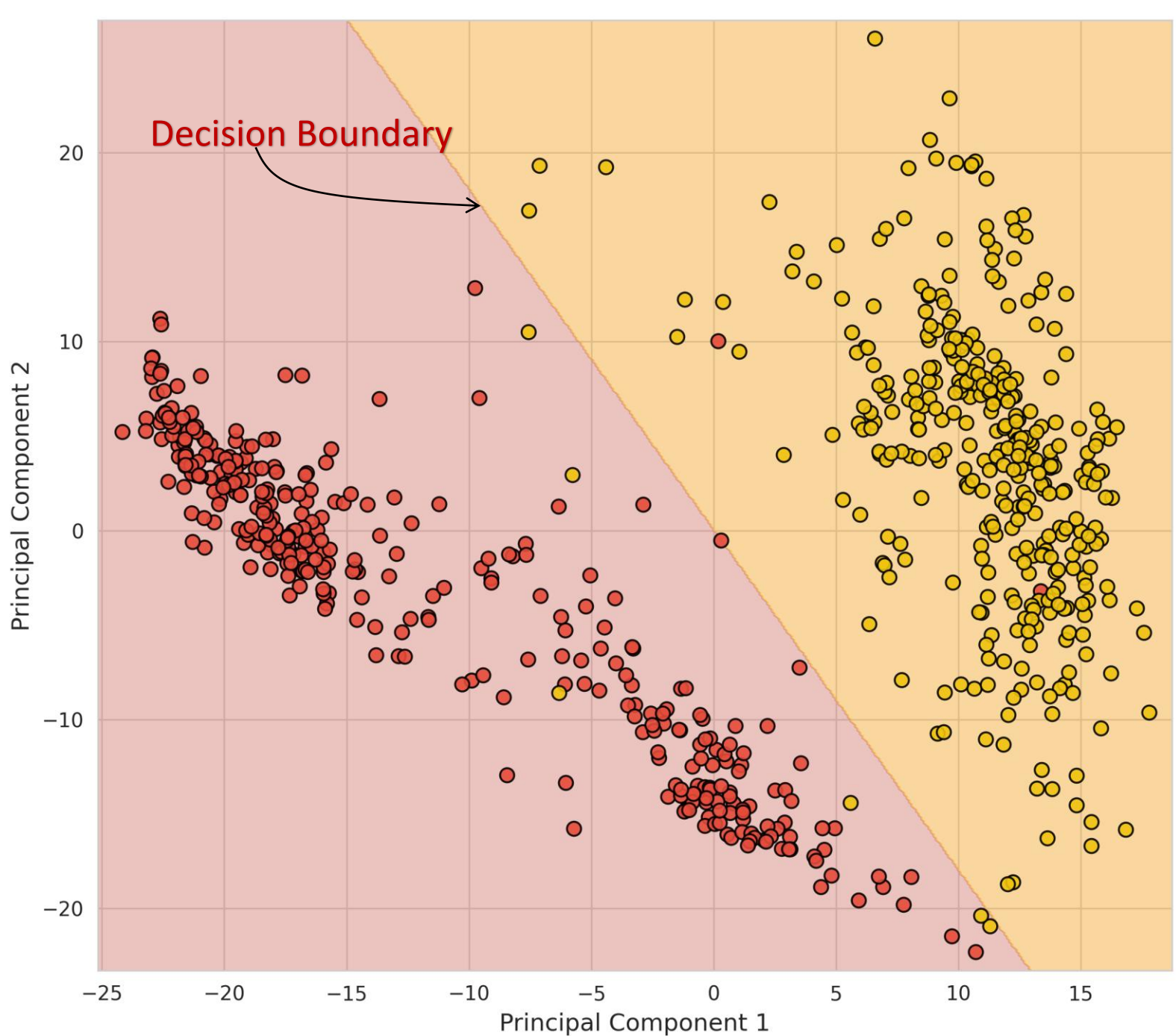


We extract the higher dimensional embedding here

- We have fine-tuned the **BLOOM-560m** model for our downstream task of classifying Tamil texts under the categories of "Inequality", "Offensive" and "Neutral".

- Trained on formal Tamil texts, **1 069** samples total size

Model Decision Boundaries:
Separating Inequality and Offensiveness



- We **reduce the higher dimensional space to 2D space** by selecting **top two principal components**, where each text sits somewhere in the 2D space.

- Then we trained **support vector machine** algorithm on the **reduced 2D space** to see if it could find a **clear decision boundary**.

Optimal separating hyperplane:

$$(0.2804 PC_1) + (0.1555 PC_2) - 0.0089 = 0$$

- The visualization demonstrates that 'Inequality' and 'Offensiveness' inhabit **largely distinct semantic spaces within Tamil discourse**, allowing the model to mathematically separate them.

Research Questions

1. Can Socio-Linguistic Harm be modeled as a multidimensional latent construct composed of representational and expressive dimensions?
2. How can Representational and Expressive harm be computationally quantified in monolingual text?
3. How can representational harm arising from the erasure of socio-historical and inequality markers be identified and quantified across cross-lingual discourse?

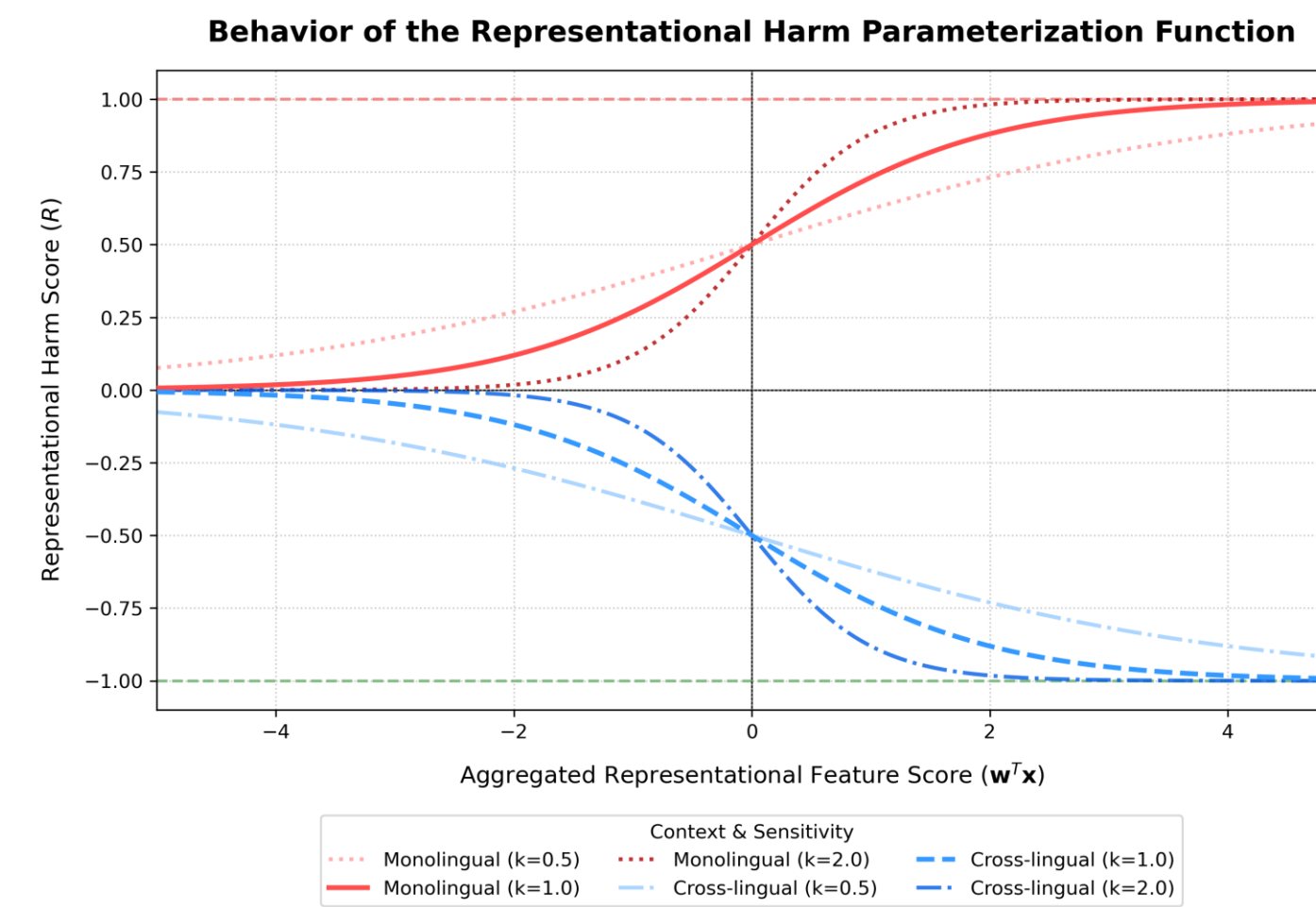
Methodology

Representational Harm: Language that promotes negative stereotypes, unequal portrayals, or erases socio-historical context of marginalized groups [3,4].

$$x(t) = [x_{hand}(t) || x_{emb}(t)]$$

$$x_{hand}(t) = \begin{bmatrix} stereotype_score(t) \\ inequality_score(t) \\ target_group_attack_score(t) \\ negative_sentiment_score(t) \\ socio - historic_erasure_score(t) \\ caste_gender_intersection_score(t) \end{bmatrix}$$

$$x_{emb}(t) = LaBSE(t); \quad z = w^T x(t); \quad f(z) = \frac{1}{1 + e^{-kz}}$$



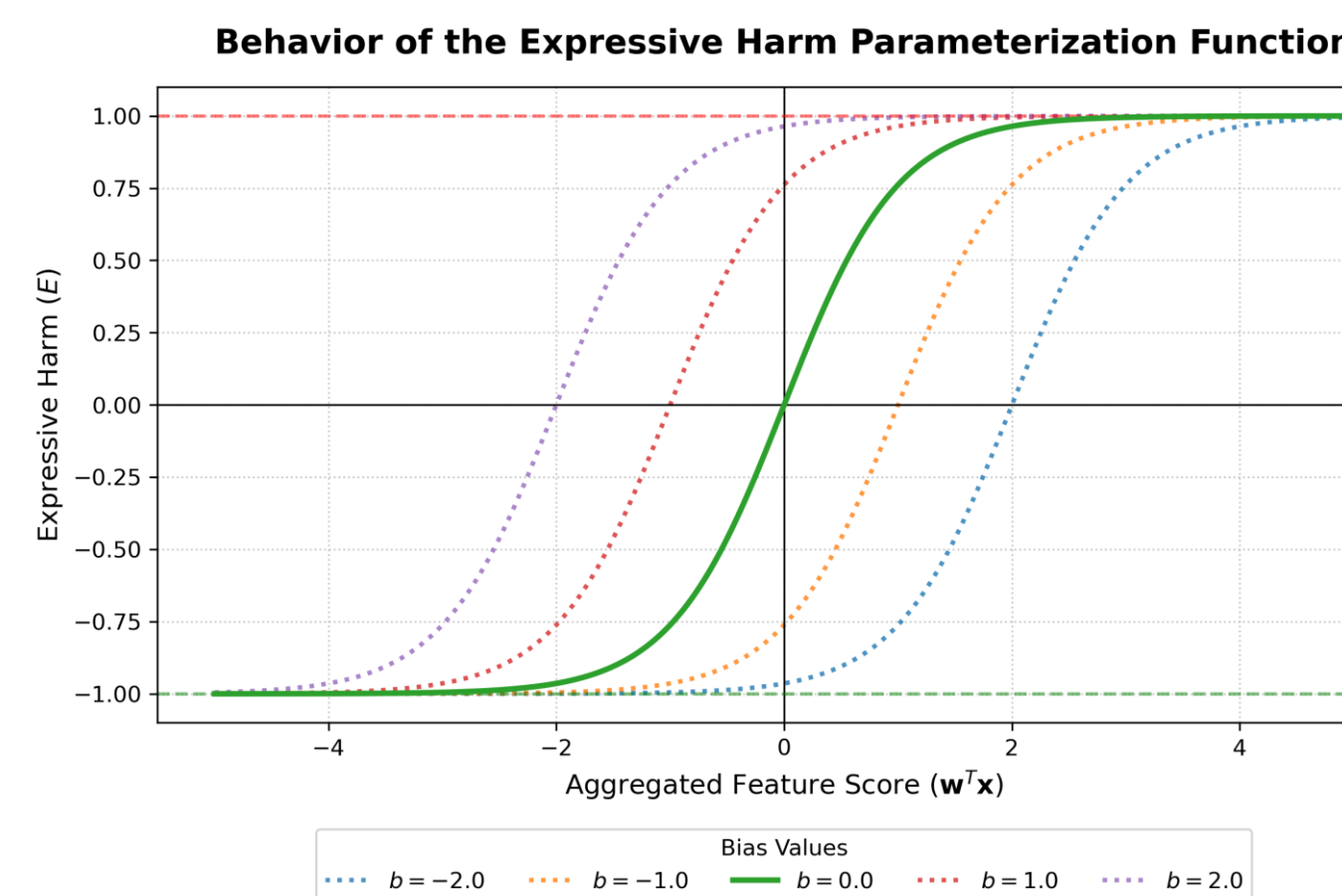
$$R_{(M,z)} = \begin{cases} f(z), & \text{if } M = \text{monolingual} \\ -f(z), & \text{if } M = \text{cross lingual} \end{cases}$$

Expressive Harm: The degree to which language conveys offense, derogation, or undermining, insulting intent toward individuals or groups [4,5].

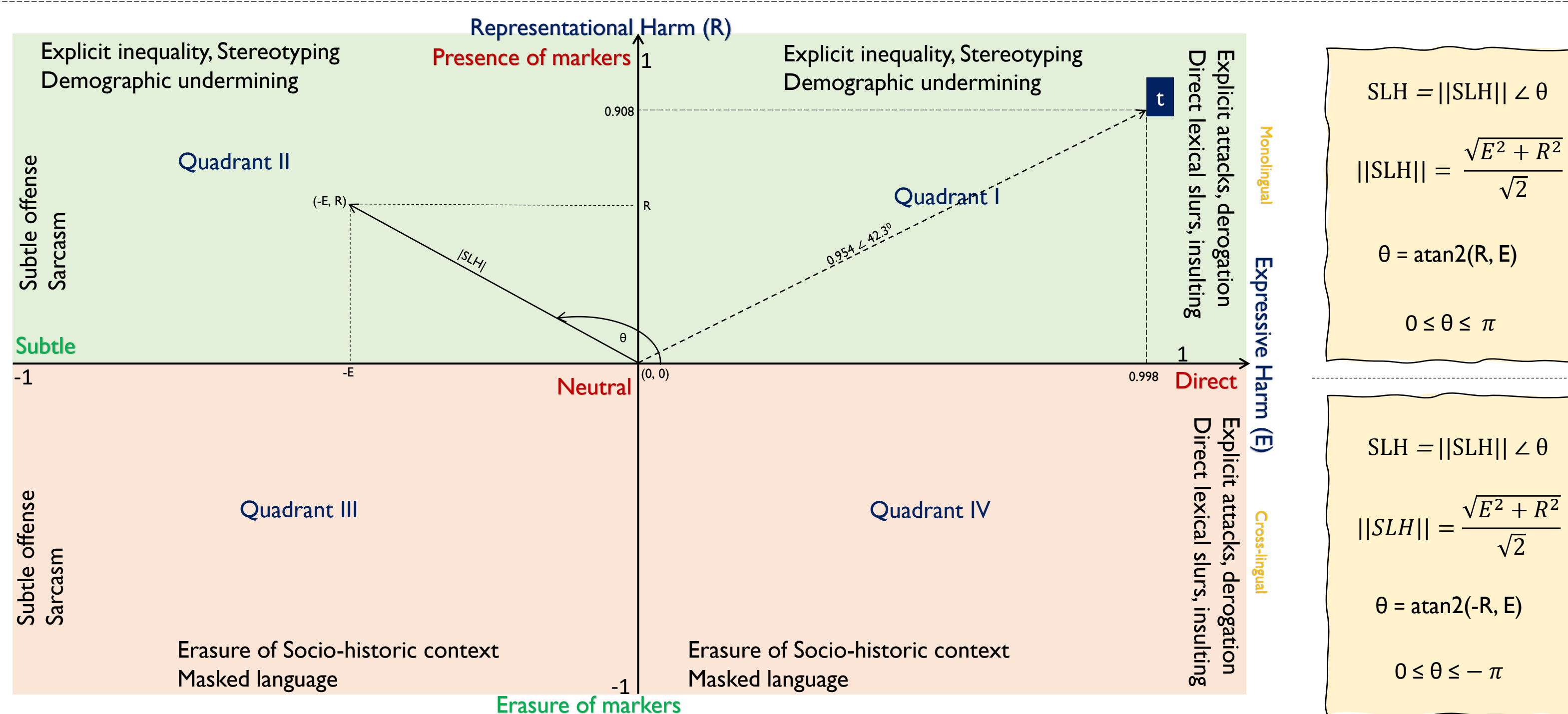
$$x(t) = [x_{hand}(t) || x_{emb}(t)]$$

$$x_{hand}(t) = \begin{bmatrix} abusive_score(t) \\ amplifier_score(t) \\ sarcastic_score(t) \\ targeted_attack_score(t) \\ threat_score(t) \\ repetition_score(t) \\ emphasis_score(t) \end{bmatrix}$$

$$x_{emb}(t) = LaBSE(t);$$



$$E(t) = \tanh(w^T x(t) + b)$$



t = “அகதியாக வந்த இந்த வந்தேறிகள் , நமது வரலாற்றுச் சிறப்புமிக்க நிலங்களில் சட்டவிரோதமாகக் குடியேறி, அதன் புனிதத்தன்மையைக் கெடுப்பதுடன் நாட்டின் அமைதிக்கும் பெரும் சவாலாக உள்ளனர்; இவர்களை உடனடியாக இங்கிருந்து அடித்து விரட்ட வேண்டும்”.

Translation: “These interlopers, who came as refugees, have illegally settled in our historically significant lands. In addition to desecrating their sanctity, they pose a major threat to the peace of the country; they should be immediately beaten and driven out from here”.

Computing Expressive Harm:

$$x(t) = [0.9, 0.7, 0.0, 1.0, 0.9, 0.0, 0.6]^T \\ w = [1.0, 0.5, 0.0, 1.0, 1.0, 0.0, 0.5]^T \\ w^T x(t) = 3.45$$

Assuming $b = 0$

$$E(t) = \tanh(w^T x(t) + b) = 0.998$$

Computing Representational Harm:

$$x(t) = [0.8, 0.9, 1.0, 0.9, 0.0, 0.0]^T \\ w = [0.5, 0.5, 1.0, 0.5, 0.0, 0.0]^T \\ z = w^T x(t) = 2.30$$

With $k = 1.0$

$$R_{(M,z)} = \frac{1}{1 + e^{-2.3}} = 0.908$$

Socio-Linguistic Harm:

$$(E, R) = (0.998, 0.908)$$

$$||SLH|| = 0.954$$

$$\theta = 42.3^\circ$$

$$SLH = 0.954 \angle 42.3^\circ$$

Preliminary Experiment

To measure the divergence of information content in each text with respect to a reference text, we apply **Kullback-Leibler** divergence.

Here P is the reference distribution.
 Q_1, Q_2 are two different texts and they are the target distribution.

$$D(P \parallel Q_1) - D(P \parallel Q_2) = \sum_{x \in X} P(x) [\log(P(x)) - \log(Q_1(x))] - \sum_{x \in X} P(x) [\log(P(x)) - \log(Q_2(x))] \\ = \sum_{x \in X} P(x) [\log(Q_2(x)) - \log(Q_1(x))]$$

By definition of “Expected Value” the following expected log-likelihood ratio holds:

$$D(P \parallel Q_1) - D(P \parallel Q_2) = E_{x \sim P} \left[\log \frac{Q_2(x)}{Q_1(x)} \right]$$

$$D(P \parallel Q_1) = \sum_{x \in X} P(x) (\log(P(x)) - \log(Q_1(x)))$$

$$D(P \parallel Q_2) = \sum_{x \in X} P(x) (\log(P(x)) - \log(Q_2(x)))$$

Then we derived the expected log-likelihood ratio to compute the relative divergence of text Q_1 from Q_2 with respect to the assumed ground truth reference P .

NOTE:
We used the real news paper articles but in this poster we chose not to explicitly mention the news paper source and article due to ethical and moral considerations.

Q1, translation: AVAILABLE

Q2, translation: AVAILABLE

P, reference: AVAILABLE

Entities Identified in Reference (P):

- [GPE]: Geopolitical Entity
- [NORP]: Nationalities, Religious groups, and Political groups.

Entities Identified in (Q2)

- [GPE]: MATCHED
- [NORP]: MATCHED

- Entities Identified in (Q1):
- [GPE]: MATCHED
 - [NORP]: UNMATCHED

Metric	Comparison	Outcome (nats)
$D(P \parallel Q_1)$	Reference (P) vs. [REDACTED] Translated (Q1)	5.3812
$D(P \parallel Q_2)$	Reference (P) vs. [REDACTED] Translated (Q2)	0.3298
Relative Divergence	$E_{x \sim P} \left[\log \frac{Q_2(x)}{Q_1(x)} \right] (> 0)$	5.0514

- Relative Divergence is measured in **natural units (nats)** in base e.

- **In numbers:** Text Q1 diverges from the original information content **156.24 times more** than Q2 with respect to the reference text P.

Conclusion and Discussion point

- The proposed Socio-Linguistic Harm (SLH) framework moves beyond monolithic "hate vs. non-hate" labels by modeling linguistic harm as a continuous, two-dimensional geometric latent construct.
- By separating Representational Harm (R) and Expressive Harm (E), the polar metric ($SLH = ||SLH|| \angle \theta$) effectively quantifies both overt explicit attacks and the subtle, systemic erasure of socio-historic markers.
- D: What are your thoughts on establishing ground truth when multiple news sources present biased or conflicting narratives?

References
[1] S. Zaheri, J. Levasseur, and C. Ahn, "Toxic comment classification," SMU Data Science Review, vol. 3, no. 1, p. 4, 2020.
[2] J. W. Du Bois, "The stance triangle," Stancetaking in discourse: Expression, evaluation, and positioning, vol. 164, no. 1, pp. 139-182, 2007.
[3] B. R. Chakravarthi et al., "Detecting caste and migration hate speech in low-resource tamil language," Language Resources and Evaluation, vol. 59, no. 3, pp. 3051-3086, 2025.
[4] Gallegos, Isabel O., et al. "Bias and fairness in large language models:A survey." Computational linguistics 50.3 (2024): 1097-1179.
[5] Blodgett, Su Lin, et al. "Language (technology) is power: A critical survey of "bias" in NLP." Proceedings of the 58th annual meeting of the association for computational linguistics. 2020.